# ULUSLARARASI GIDA, TARIM VE HAYVANCILIK KONGRESİ

19-22 EYLÜL 2019
GAZİANTEP

# KONGRE TAM METİN KİTABI

Editörler
Prof. Dr. Osman ERKMEN
Nurlan AKHMETOV

İSBN 978-605-7811-08-0
İKSAD YAYINEVİ

# TAM METİN KİTABI

# *ULUSLARARASI GIDA, TARIM VE HAYVANCILIK KONGRESİ*

## 19-22 EYLÜL 2019
## GAZİANTEP, TÜRKİYE


**Editörler**

**Prof. Dr. Osman ERKMEN**

**Nurlan AKHMETOV**



**Institute Of Economic Development And Social Researches Publications®**

(The Licence Number of Publicator: 2014/31220)

TURKEY

TR: +90 342 606 06 75

E posta: kongreiksad@gmail.com

www.iksad.org www.iksadkongre.org



Bu kitabın tüm hakları İKSAD'a aittir. İzinsiz kopyalanamaz ve çoğaltılamaz.

Kitapta bulunan eserlerin yasal ve etik sorumluluğu yazarlara aittir.

**İksad Publications - 2019©**



**Yayın Tarihi: 10.10.2019**

**ISBN – 978-605-7811-08-0**

# KONGRE KÜNYESİ

## KONGRE ADI

**ULUSLARARASI GIDA, TARIM VE HAYVANCILIK KONGRESİ**

## TARİHİ VE YERİ

19-22 Eylül 2019, Gaziantep, Türkiye

## DÜZENLEYEN KURUM

İKSAD- İktisadi Kalkınma ve Sosyal Araştırmalar Enstitüsü

## KONGRE BAŞKANI

Prof. Dr. Osman ERKMEN

## KOORDİNATÖR

Nurlan AKHMETOV

## YABANCI KONUŞMACILAR / KATILIMCILAR

Mohamed Nadjib BOUKHATEM (ABD)
Siham BAHLOULI (Cezayır)
Abdalla Abdalrazig Abdalla HUSSEIN (Sudan)
A. GAVRILOVA, A. YOSIFOV (Rusya)
Falah Saleh MOHAMMED (Irak)
Guguli DUMBADZE (Gürcistan)

## KONGRE DİLLERİ

Türkçe, İngilizce, Rusça, Arapça

## SUNUM ŞEKLİ

Sözlü Sunum

## BİLİM VE DANIŞMA KURULU


Prof. Dr. Tuncay GÜMÜŞ,

Namık Kemal University

Prof. Dr. Şenol İBANOĞLU,

Gaziantep University

Prof. Dr. Sibel FADILOĞLU,

Gaziantep University

Prof. Dr. Sevim KAYA,

Gaziantep University

Prof. Dr. Mutlu B. Akın, Harran University

Prof. Dr. Mustafa BAYRAM,

Gaziantep University

Prof. Dr. Muammer Arıcı,

Yıldız Teknik University

Prof. Dr. Mihriban KORUKLUOGLU,
Uludağ University

Prof. Dr. Medeni MASKAN,

Gaziantep University

Prof. Dr. Hüseyin BOZKURT,

Gaziantep University

Prof. Dr. Hamit KÖKSEL,

Hacettepe University

Prof. Dr. Hasan YETİM,

İstanbul S. Zaim University

Prof. Dr. F. Yeşim EKİNCİ,

Yeditepe University

Prof. Dr. Akbar VALADBIGI,

Elmi Korbordi University

Prof. Dr. Mustafa BAYRAM,

Gaziantep University

Prof. Dr. Esra İBANOĞLU,

Gaziantep University

Prof. Dr. Ebru SENEL,

Ankara University

Prof. Dr. Altunay PERENDECİ,

Akdeniz University

Prof. Dr. Saodat DOVLATOVA,

Tashkent University

Prof. Dr. Ali Rıza TEKİN,

Gaziantep University

Prof. Dr. Zafer GÖNÜLALAN,

Erciyes University

Assoc. Prof. Dr. Abdullah DİKİCİ ,

Uşak University

Assoc. Prof. Dr. Serap COŞANSU
AKDEMİR, Sakarya University

Assoc. Prof. Dr. Ayşe Handan BAYSAL

Assoc. Prof. Dr.K. Bülent BELİBAĞLI,
Gaziantep University

Assoc. Prof. Dr.Hacı Ali GÜLEÇ,

Trakya University

Assoc. Prof. Dr.EMİNE (ALBEN)
ERÇELEBİ, Gaziantep University

Assoc. Prof. Dr. Derya KOÇAK YANIK,
Gaziantep University

Assoc. Prof. Dr. Berna SARIYAR
AKBULUT, Marmara University

Assoc. Prof. Dr.Çiğdem AYKAÇ SOYSAL,
Gaziantep University

Assoc. Prof. Dr. Stephan
MARCHENKO, Komrad State Univeristy

Assoc. Prof. Dr.Ömer ŞİMŞEK,

Pamukkale University

Assoc. Prof. Dr.Mustafa Türker,

Pak Gıda Üretim Pazarlama A.Ş.

Assoc. Prof. Dr. Arzu Çağrı Mehmetoğlu,
Sakarya University

Assoc. Prof. Dr.A. Coşkun DALGIÇ,
Gaziantep University

Assist. Prof. Dr. Gülten Şekeroğlu,
Gaziantep University

Dr. Almaz AHMETOV,

Azerbaijan Medical University

Dr. Aytül SOFU,

Süleyman Demirel University

Dr. Abdullah KURT,

Bitlis Eren University

Dr. Hidayet SAĞLAM,

Kilis 7 Aralık University

Dr. Funda KARBANCIOĞLU GÜLE,
İstanbul Teknik University

Dr. H. İmge OKTAY BAŞEĞMEZ,

Adana Alparslan Türkeş University

Dr. Bauryzhan BOTAKARAYEV,

Akhmet Yassawi University

Dr. Havva MEHTIEVA,

Moscow State University

Dr. Cem Okan ÖZER, Nevsehir Hacı
Bektas Veli University

Dr. Hasene KESKİN ÇAVDAR,

Gaziantep University

Dr. Hicran UZUN KARKA,

Gaziantep University

M. Şamil ARGUN,

Bitlis Eren University

Dr. Necla ÖZDEMİR,

Bitlis Eren University

Dr. Serdar AYDEMİR, Enka Süt ve Gıda
Mamulleri Sanayi ve Ticaret Anonim
Şirketi

# MİKSOMİSETLERİN MORFOLOJİK YAPILARI; PLASMODYUM TİPLERİ

MORPHOLOGICAL STRUCTURE OF MYXOMYCETES; THE PLASMODIUM TYPES

**Hayri BABA**

Hatay Mustafa Kemal University, Faculty of Science and Literature, Department of Biology, Hatay

**Mustafa SEVİNDİK**

Osmaniye Korkut Ata University, Bahçe Vocational School, Department of Food Processing, Osmaniye

**Celal BAL**

Gaziantep University, Oğuzeli Vocational School, Gaziantep

**Hasan AKGÜL**

Akdeniz University, Faculty of Science, Department of Biology, Antalya

## ÖZET

Miksomisetlerin biyolojisi spordan başlayarak yaşam döngüsündeki aşamaları izler. Spor çimlenerek bir miksamoeba veya bir swarm cell'e dönüşmeye başlar, olumsuz şartlar altında bir mikrosist oluşturur veya optimal şartlarda sonunda plazmodyuma gelişir. Plasmodium, sporların oluştuğu fruktifikasyon verecek olan vejetatif aşamadır. Plasmodyum, duvarsız (genel olarak sümüksü) serbest yaşayan bir çok çekirdekli, yaş ve türlerine göre boyut ve morfolojik ayrıntılara göre değişen hareketli protoplazma kütlesidir. Bakteriler, mantarlar ve karşılaşabilecekleri diğer organik maddeler ile beslenirler. Miksomisetlerin en belirgin aşaması, asimilatif yapı olan plasmodyumdur. Üç ana plasmodium tipi vardır,

Phaneroplasmodium sıkça görülen en büyük, en göze çarpan ve renkli plazmodiyal tipdir. Tipik olarak, geniş bir alanı kapsayabilen birçok fruktifikasyon verebilecek kadar geniş gövdeye sahiptir. Olgun faneroplasmodium, içinde akış protoplazmasının kanallarının ortaya çıktığı, ön fan şeklinde bir pigmentli granüler protoplazma tabakasına sahiptir. Bu kanallar önden arkaya doğru yönlendirilir, daha kısa bağlantı kanalları dik açıda veya ana kanallara çapraz olarak yönlendirilir, ön kanallar plasmodyumun nispeten kalın iplikçiklerin arka retikulumuyla birleşir. Physarales takımının üyelerinde bu tip plazmodyum görülür.

Aphanoplasmodium Stemonitales ve Ceratiomyxales takımlarının karakteristik plasmodyum tipidir, erken gelişim evrelerinde, kutupsallık ve yön hareketi olmayan düzleştirilmiş, iplik benzeri, saydam, neredeyse görünmeyen ipler ağıyla karakterize edilir. Bu erken evreler, farklı ektoplazmik ve endoplazmik bölgelere sahip değildir ve akış protoplazması, kaba granül değildir. Aphanoplasmodium görünmez anlamında olup doğada çok nadir görülebilen bir plazmodyum tipidir. Damarları müsilajımsı bir kından yoksundur ve çok incedir.

Protoplasmodium varlığı boyunca mikroskobik kalan en küçük en ilkel plazmodium tipidir. Oldukça granüler ve homojen protoplazma, diğer plasmodial tiplerde vasküler benzeri retikülat kordonları ve tipik gelişmiş fanları geliştiremeyen plaka benzeri bir şekle sahiptir, damarları yoktur, düzensiz ve yavaş bir protoplazmik akış sergiler. Her plazmodyum, Echinosteliales türlerinde ve bazı Liceales türlerinde olduğu gibi tek bir küçük sporangiuma dönüşür. Diğer iki plazmodiyal tipten farklı olarak, retiküle edilmiş bir plazmodyum üretmez veya kendisiyle veya başka bir plazmodtyumla birleşmez. Bir üst sınıra ulaşıldığında, protoplasmodium iki küçük plazmodyumu üretmek için ayrılır ve sporülasyonun ardından, her protoplasmodyum tek bir sporangium üretir.

Belki de dördüncü tip bir plasmodium Trichiaceous plasmodium, hem aphano hem de phaneroplasmodium'un özelliklerine sahip bir kombinasyondur. Gelişimi için serbest su

gerektiren, genel olarak belirgin bir antero-posterior polariteye sahip olan Trichiales ordosu üyelerinde görülen bir plasmodyum tipidir.

**Anahtar Kelimeler**: Miksomiset, Morfolojik yapı, Plasmodyum Tipleri

**ABSTRACT**

The biology of the myxomycetes narrative follows the stages in the life cycle beginning with the spore. The spore germinates giving rise to either a myxamoeba or a swarm cell that may form a microcyst under adverse conditions, or given optimal conditions, may fuse with genetically compatible types, eventually developing into a plasmodium. The plasmodium is the stage that develops into the fruiting bodies where spores are formed. The plasmodium is a wall-less (generally slime covered) free-living multinucleate, motile mass of protoplasm which varies in size and morphological detail with age and species. They feed on bacteria, fungi, and any other organic matter that is encountered which they can engulf. The single most distinctive stage of a myxomycete is the assimilative structure, the plasmodium.

There are three major types of plasmodia;

Phaneroplasmodium is the largest, most conspicuous, and colorful plasmodial type frequently seen in the field, Phaneroplasmodia are the most common and the most variable of the plasmodial types. It typically gives rise to many fruiting bodies that may cover an extensive area. The mature phaneroplasmodium has an anterior fan-shaped sheet of pigmented granular protoplasm, within which channels of streaming protoplasm originate. These channels are orientated from front to back, with shorter connecting channels at right angles and or diagonally to the main channels, with the anterior channels merging into the plasmodium's posterior reticulum of relatively thick strands This type of plasmodium grows best under drier conditions where free water is absent. Members of the order Physarales are the best examples of this type of plasmodium. The phaneroplasmodium is characteristic of the Physarales, Trichiales, and most if not all of the Liceales.

Aphanoplasmodium is characteristic of the Stemonitales and Ceratiomyxales, characterized in its early stages of development by a network of flattened, thread-like, transparent, almost invisible strands that lack polarity and directional movement. These early stages lack distinct ectoplasmic and endoplasmic regions, and the streaming protoplasm is not coarsely granular.

Protoplasmodium is smallest of the plasmodial types that remains microscopic throughout its existence. The highly granular and homogeneous protoplasm has a plate-like shape that fails to develop vein-like reticulate strands and advancing fans typical of other plasmodial types. Each plasmodium gives rise to a single, tiny sporangium, as in species of Echinostelium. protoplasmodium with the Echinosteliales and with some Liceales. Unlike the other two plasmodial types it does not produce a reticulated plasmodium or fuse with either itself or any other plasmodium. Upon reaching an upper size limit the protoplasmodium undergoes binary plasmotomy to produce two daughter plasmodia, and upon sporulation, each protoplasmodium produces a single sporangium.

Perhaps a fourth type is a combination of the first two as in the. The Trichiaceous plasmodium has characteristics of both the aphano- and phaneroplasmodium, requiring free water for development, having a generally appressed appearance and distinct antero-posterior polarity.

**Keywords:** Myxomycetes, Morphological Structure, Fruiting Bodies

## INTRODUCTION

The myxomycetes (plasmodial slime molds or myxogastrids) are a group of eukaryotic microorganisms that form part of the supergroup Amoebozoa (Adl et al., 2012). The life cycle involves two trophic stages one consisting of microscopic uninucleate amoeboflagellate cells and the other of a multinucleate plasmodium along with a reproductive stage; fruiting body or sporocarp (Rojas et al.,2017; Sevindik et al., 2018). A fruiting body develops from plasmodium.

Myxomycetes have long been known to occur on decaying wood, litter, soil, herbivore dung and the bark surface of living trees (Stephenson and Stempen, 1994; Baba et al., 2018; Ergül and Akgül, 2011). Myxomycetes have been recorded from every major type of terrestrial ecosystem examined to date, and at least a few species have been recovered from aquatic habitats (Lindley et al., 2007; Baba and Sevindik, 2018). Temperature and moisture are thought to be the main factors limiting the occurrence of myxomycetes in nature (Alexopoulos, 1963; Ergül et al., 2005a; Ergül et al., 2005b). Species richness tends to increase with increasing diversity and biomass of the vascular plants providing the resources that support the bacteria and other microorganisms upon which the two trophic stages in the myxomycete life cycle feed (Madelin, 1984; Stephenson, 1989; Ergül et al., 2016).

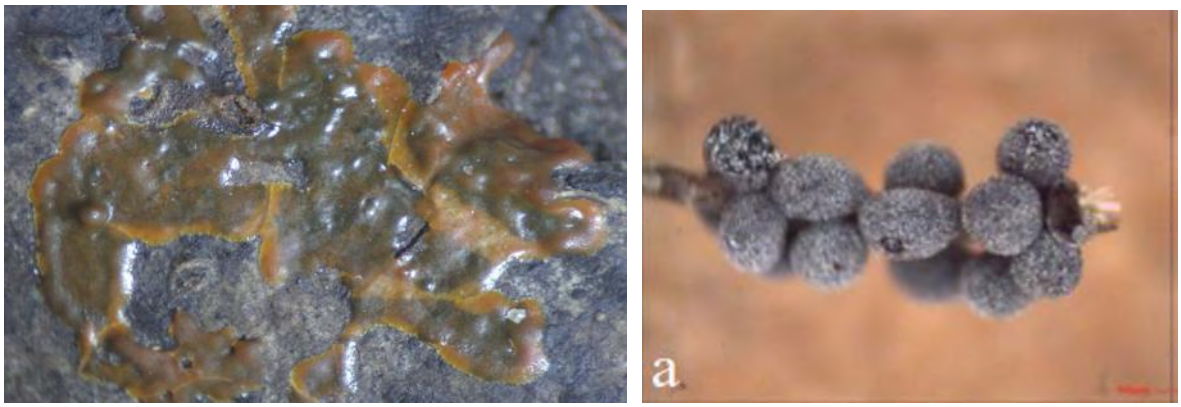
The plasmodium is a membrane-bound single cell containing multiple nuclei. The plasmodium of some species may have a slime sheath, which appears as a "slime track" left behind when the plasmodium migrates across a given substrate. Plasmodial color may be affected by pH, temperature or ingested materials. Plasmodium as the main trophic stage in their developmental cycle. In nature, plasmodia live in dark and humid places: under the bark of rotten logs, under decaying leaves. Then they migrate to the surface of the substrata where after some time they sporulate. In a standard laboratory culture plasmodia are protected from light and kept in the dark because radiation has an inhibitory effect on their growth. Plasmodia of many species are pigmented, the colour being widely differentiated and ranging from white, yellow, orange, red of various shades, to black. Very frequently taxonomists find in nature the same species of slime mould with different plasmodia colours, and changes in pigmentation have also been observed in plasmodia which stayed for some period on the surface of the substrata (Rakoczy, 1998).

Under unfavorable conditions (low temperatures, low moisture, overcrowding, lack of food, or accumulation of metabolic products), the plasmodium will either migrate to a new location or desiccate and harden into a durable resting or dormant state that can survive for seven months or longer under controlled conditions (Alexopoulos, 1960; Gray and Alexopoulos, 1968). This dormant stage is called the sclerotium and is composed of spherules that are contiguous and enclose one to many nuclei. Once favorable conditions return, the sclerotium will hydrate and the active plasmodium will crawl out, continue feeding, and produce fruiting bodies.

## PLASMODIAL TYPES

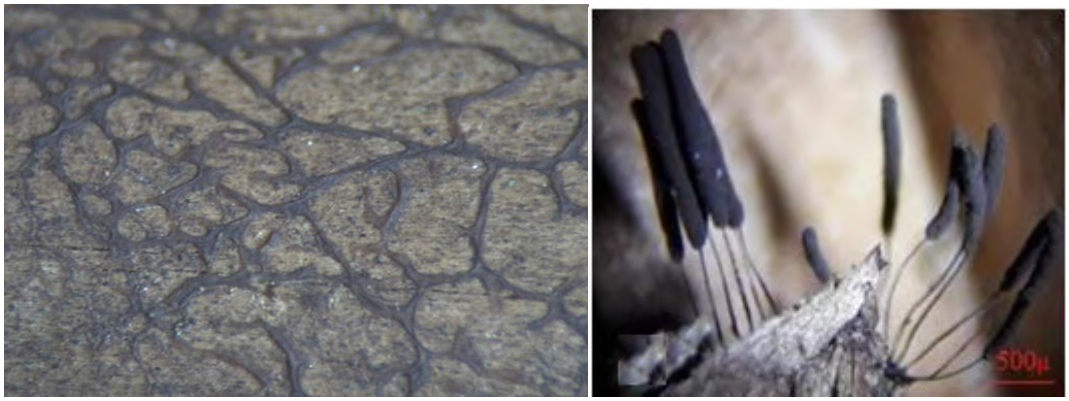
There are four distinct morphological forms of plasmodia, the phaneroplasmodium, aphanoplasmodium, protoplasmodium, and an intermediate type called the trichiaceous plasmodium (Gray and Alexopolous, 1968), but the most common is the phaneroplasmodium.

- The phaneroplasmodium (Greek phaneros = visible) typically has a fan-like plasmodium that exhibits polarity and reversible protoplasmic streaming, and can form hundreds of fruiting bodies or one large aethalium up to 70 cm across (Keller and Braun, 1999). The phaneroplasmodium has a broad, anterior feeding edge, followed by a defined set of reticulate, posterior trailing veins. This type appears as a fan-shaped network of veins containing streaming protoplasm. This type is common in the order Physarales.



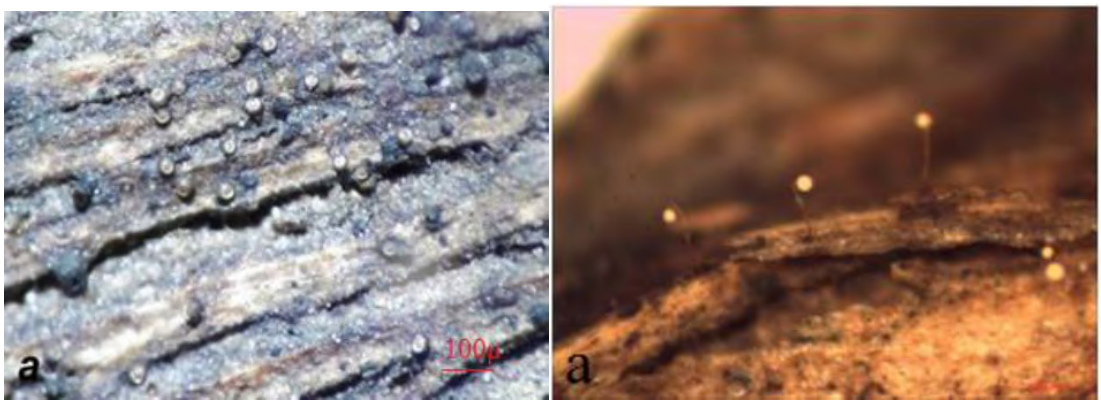

The Phaneroplasmodium type and *Physarum confertum* (Baba, 2008)

- The aphanoplasmodium (Greek aphanes = invisible) requires the presence of free water, forms a vein-like reticulate pattern that lacks distinct anteroposterior polarity, and exhibits erratic protoplasmic streaming. The characteristically thin and hyaline aphanoplasmodium is adapted to growing in the interstices of wood, flowing through the phloem and xylem (Gray and Alexopoulos, 1968). A second type, the aphanoplasmodium is rarely observed in nature. The veins are very thin and a slime sheath is lacking. Aphanoplasmodia are characteristic of the order Stemonitales.




The Aphanoplasmodium type and *Stemonitopsis amoena* (Baba and Tamer, 2008)

- The protoplasmodium (Greek protos = first) is microscopically rounded or irregular-shaped, does not develop the vein-like network characteristic of the other plasmodial types, and will give rise to only a single stalked sporangium. Protoplasmodium is the third and probably the most primitive type. Always microscopic, this plasmodium has no veins; exhibits slow, irregular protoplasmic streaming and produces a single, very small fruiting body. Protoplasmodia are found in the Echinosteliales and Liceales.



The Protoplasmodium types *Licea kleistobolus* and *Echinostelium minutum* (Baba and Tamer, 2008)

- The Trichiaceous plasmodium has characteristics of both the aphano and phaneroplasmodium, requiring free water for development, having a generally appressed appearance and distinct antero-posterior polarity and occurs in the order Trichiales.

The Trichiaceous plasmodium type *Arcyria incarnata* and *Trichia decipiens* (Baba and Tamer, 2008).

## CONCLUSION

The life cycle of myxomycetes involves three morphologically distinct stages. These are the amoeboflagellate stage, the plasmodium and the fruiting body containing spores. The plasmodium; single most distinctive stage of a myxomycete is the assimilative structure, feeds on bacteria, yeast, and other microorganisms, by phagocytosis and fusions with other plasmodia. The plasmodium is membrane-bound, acellular, and naked.

## REFERENCES

Adl S, Simpson A, Lane C, Lukes J, Bass D, Bowser S, Brown M, Burki F, Dunthorn M, Hampl V, Heiss A, Hoppenrath M, Lara E, Gall L, Lynn D, McManus H, Mitchell E, Muzky-Stanridge S, Parfrey L, Pawlowski J, Rueckert S, Shadwick L, Schoch C, Smirnov A, Spiegel F. 2012 – Classification of Eukaryotes. Journal of Eukaryotic Microbiology 59, 429–493.

Alexopoulos CJ. 1960. Gross morphology of the plasmodium and its possible significance in the relationships among the Myxomycetes. Mycologia 52, 1–20.

Alexopoulos CJ. 1963. The Myxomycetes. II. Botanical Review 29, 1–79.

Baba H, 2008. A New Myxomycetes Genus and three species record for Turkey. International Journal of Botany 4: 336-339.

Baba, H. and Tamer, A.Ü. 2008. A New Myxomycetes Genus and Three Species Record for Turkish mycoflora. The Herb Journal of Systematic Botany, 15(2), 81-86.

Baba, H. and Sevindik, M. (2018). The roles of myxomycetes in ecosystems. J Bacteriol Mycol Open Access, 6(3), 165-166.

Baba, H., Gelen, M., & Sevindik, M. (2018). Taxonomic investigation of myxomycetes in Altınözü, Turkey. Mycopath, 16(1), 23-31

Ergul, C. C., Akgul, H., & Oran, R. B. (2016). New records of Mycetozoa taxa from Turkey. Oxidation Communications, 39(2), 1615-1623.

Ergul, C. C., Dulger, B., & Akgul, H. (2005). Myxomycetes of Mezit stream valley of Turkey. Mycotaxon.

Ergul, C. C., Dulger, B., Oran, R. B., & Akgul, H. (2005). Myxomycetes of the western Black Sea Region of Turkey.[Erratum: 2006 Apr-June, v. 96, p. 362.]. Mycotaxon.

Ergül, C. C., & Akgül, H. (2011). Myxomycete diversity of Uludağ national park, Turkey. Mycotaxon, 116(479), 1-16.

Gray WD, Alexopoulos CJ. 1968. Biology of the Myxomycetes. Ronald Press, New York.

Keller, H.W., Braun, K.L., 1999. Myxomycetes of Ohio: Their Systematics, Biology, and Use in Teaching (Ohio Biological Survey Bulletin New Series, Vol. 13, No. 2). Ohio Biological Survey, Columbus, OH.

Lindley, L.A., Stephenson, S.L., Spiegel, F.W., 2007. Protostelids and myxomycetes isolated from aquatic habitats. Mycologia 99, 504–509.

Madelin, M.F., 1984. Myxomycete data of ecological significance. Trans. Br. Mycol. Soc. 83, 1–19.

Rakoczy L. 1998. Plasmodial Pigmentation of the Acellular Slime Mould Physarum polycephalum in Relation to the Irradiation Period, Polish Journal of Environmental Studies Vol. 7, No. 6, 337-342

Rojas C, Rollins A.W. and Stephenson S.L. 2017. Distribution of Myxomycetes Among the Microhabitats Available for these organisms in Tropical Forests Myxomycetes. Biology, Systematics, Biogeography, and Ecology. Academic Press. Elsevier. Pp. 205-251.

Sevindik, M., Baba, H., Bal, C., Colak, O. F., & Akgul, H. (2018). Antioxidant, Oxidant and Antimicrobial Capacities of Physarum album. *Journal of Bacteriology & Mycology: Open Access*, *6*(6), 317-320.

Stephenson, S.L., 1989. Distribution and ecology of myxomycetes in temperate forests. II. Patterns of occurence on bark surface of living trees, leaf litter, and dung. Mycologia 81, 608–621.

Stephenson, S.L., Stempen, H., 1994. Myxomycetes: A Handbook of Slime Molds. Timber Press, Portland.